

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester I Examination, 2024-25 (NEP)

DEPARTMENT OF MATHEMATICS**Paper Code: MATMJ-11 / MATMN-11 : Calculus**

Time: 3 Hours

Maximum Marks: 60

Instructions: Attempt FIVE questions in total. All questions carry equal marks (12 marks each).

Question 1

- (a) State and prove Leibniz's Theorem for finding the n -th derivative of the product of two functions. [6]
- (b) If $y = (x^2 - 1)^n$, prove that $(x^2 - 1)y_{n+2} + 2xy_{n+1} - n(n+1)y_n = 0$. [6]

Question 2

- (a) Find all asymptotes of the curve $x^3 + 2x^2y - xy^2 - 2y^3 + 4x^2 - y^2 + 2x + 1 = 0$. [6]
- (b) Derive the expression for radius of curvature ρ in polar coordinates $r = f(\theta)$. [6]

Question 3

- (a) Define envelope and evolute of a family of curves. Find the envelope of the family of straight lines $\frac{x}{a} + \frac{y}{b} = 1$, where the parameters a and b are connected by $a + b = c$ (c is constant). [6]
- (b) Find the coordinates of the center of curvature at any point (x, y) on the parabola $y^2 = 4ax$. [6]

Question 4 Trace the curve $y^2(a - x) = x^3$ (Cissoid of Diocles) completely, giving details of symmetry, origin, asymptotes, tangents at origin, and region of existence. [12]

Question 5

- (a) Find the points of inflexion on the curve $y = \frac{c^2x}{x^2 + a^2}$. [6]
- (b) Determine the concavity and convexity of the curve $y = x^4 - 6x^3 + 12x^2 - 8x + 1$. [6]

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